

Homework 14 Solutions

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Problem 31.1

Let (S, P) be a sample space with $S = \{1, 2, 3, \dots, 10\}$ and $P(x) = 0.1$.

- a) $A = \{2, 4, 6, 8, 10\}, P(A) = 0.5$
- b) $B = \{1, 3, 5, 7, 9\}, P(B) = 0.5$
- c) $C = \{2, 3, 5, 7\}, P(C) = 0.4$
- d) $D = \{\}, P(D) = 0$

Problem 31.3

Let (S, P) be a sample space with $S = \{HHHH, HHHT, HHHHTH, HHTT, \dots, TTTT\}$ and $P(x) = \frac{1}{16}$. Let A be the event "equal number of heads and tails".

- a) $A = \{HHTT, HTHT, HTTH, THHT, THTH, TTHH\}$
- b) $P(A) = \frac{6}{16} = \frac{3}{8}$

Problem 31.4

Let (S, P) be the sample space for flipping a fair coin 10 times. Let A be the event that we record an even number of heads and tails. Note that $P(x) = \frac{1}{1024}$.

Then $P(A) = |A| \cdot \frac{1}{1024} = \binom{10}{5} \cdot \frac{1}{1024} = 252 \cdot \frac{1}{1024} = \frac{63}{256}$

Problem 31.5

There are $\binom{n}{h}$ ways of flipping exactly h heads, and 2^n ways of flipping a coin n times. So the odds of flipping exactly h heads is

$$\binom{n}{h} \cdot \frac{1}{2^n}.$$

Problem 31.7

Let (S, P) be the sample space for rolling a pair of dice where $S = \{(1, 1), (1, 2), \dots, (6, 5), (6, 6)\}$ and $P(x) = \frac{1}{36}$.

- a) $A = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$
- b) $P(A) = \frac{5}{36}$

Problem 32.1

Let (S, P) be a sample space where $S = \{1, 2, 3, 4, 5\}$, $P(1) = 0.1$, $P(2) = 0.1$, $P(3) = 0.2$, $P(4) = 0.2$, and $P(5) = 0.4$.

- a) $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $P(A|B) = \frac{0.3}{0.5} = \frac{3}{5}$
- b) $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $P(A|B) = \frac{0.3}{0.5} = \frac{3}{5}$
- c) $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $P(A|B) = \frac{0.3}{0.5} = \frac{3}{5}$
- d) $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $P(A|B) = \frac{0.3}{0.5} = \frac{3}{5}$

Problem 32.4

Let (S, P) be the sample space for rolling a pair of dice where $S = \{(1, 1), (1, 2), \dots, (6, 5), (6, 6)\}$ and $P(x) = \frac{1}{36}$. Let A be the event that the dice sum to 7, and let B be the event that neither die shows a 2. Then $A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, $B = \{1, 3, 4, 5, 6\} \times \{1, 3, 4, 5, 6\}$, $A \cap B = \{(1, 6), (3, 4), (4, 3), (6, 1)\}$, and observe that $|B| = 25$. Then

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{4/36}{25/36} = \frac{4}{25}.$$

Problem 32.31

Let (S, P) be a sample space with $S = \{H, T\}$ and $P = \{(H, p), (T, (1 - p))\}$, let A be the event HT , and B be the event TH .

- a) $P(A) = p \cdot (1 - p)$.
- b) $P(B) = (1 - p) \cdot p = p \cdot (1 - p) = P(A)$.
- c) $P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B)} = \frac{P(A)}{2 \cdot P(A)} = \frac{1}{2}$.
- d) $P(B|A \cup B) = \frac{P(B \cap (A \cup B))}{P(A \cup B)} = \frac{P(B)}{P(A \cup B)} = \frac{P(B)}{P(A) + P(B)} = \frac{P(A)}{2 \cdot P(A)} = \frac{1}{2}$.
- e) Choose two events that have an equal number of heads and tails, like HT and TH . Then assign each of these events to their desired outcomes. Now, flip the coin generating one of the events HH, HT, TH, TT . If the generated event is not one of the desired events, flip again. If it is one of the desired events, you have your outcome.